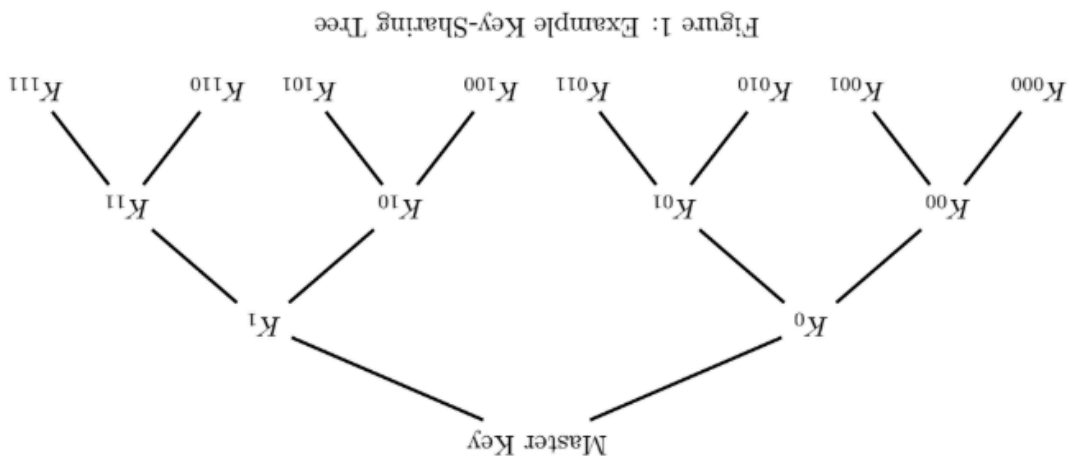
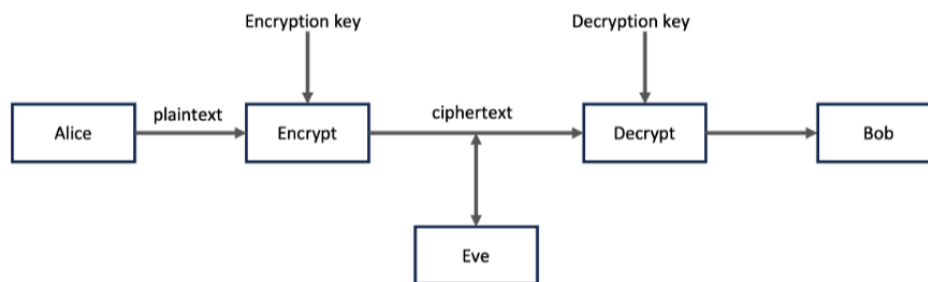


Currently we have multiple methods for cryptography which have different purposes. The primary method would be Elliptic Curves for public-key cryptography, ensuring that information sent over the communication channel is well protected. The start and end device will use Hashing like SHA-512 and Hashed Message Authentication Codes (HMACs) to ensure integrity verification of the data.

Fig. 1: Authentication Channel

Authentication goal:

- Bob wants to be sure that only Alice could have sent the message



Generated cryptographic secrets must be securely stored in nonvolatile storage to ensure data-at-rest protection. To protect the data stored by the HSM, a secure binary tree will be used to protect the PINs of each group that have permissions to read and/or write files in the HSM. A master key will be created, where each node is a derived key for the PIN for each group. HDKF is a key derivation protocol from HMAC, It is an extract and expand system, where the master key is expanded using HMAC digest, and the sub-keys are derived with the digest.

MPU:

A memory protection unit (MPU) can be utilized to enhance security, providing robust memory protection mechanisms. The MPU supports up to eight configurable memory regions, enabling granular access control. Additionally, the Sub Region Disable (SRD) feature allows for efficient use of memory regions by selectively disabling specific sub-regions. Furthermore, the MPU can enable a background region that enforces default memory map attributes, ensuring a secure and consistent memory access policy. Memory security can also be tested at compile time through tools like Valgrind to minimize the risk of memory-related vulnerabilities.

Crypto Hash:

The system leverages WolfSSL/Simple Crypto, which provides a suite of cryptographic algorithms available for implementation. Securely authenticating subscribed users presents a challenge due to the absence of a Trusted Third Party (TTP) (Fig. 1). Authentication is facilitated through an HMAC protocol that requires a cryptographic key accompanied by a user ID, password, and cryptographic signature. For secure key exchange, the Elliptic Curve key exchange protocol is employed to establish shared secrets between communicating parties. Additionally, a timekeeping protocol or system, such as the Network Time Protocol (NTP) or a Real-Time Clock (RTC), is utilized to ensure accurate timestamping of data frames, enhancing data integrity and synchronization.

How CryptoHash is implemented on SoC:

Further Security considerations were also taken when planning on implementing encryption. SHA-512 should be used when verifying the group ID as well as timestamps. In case of the subscription not being valid, the attacker could be delayed when processing the debugger. By doing so, attackers would be dismayed from attempting to brute force a way into getting access to the frames. Anti-debugger tools such as IsDebuggerPresent() would check if a debugger is being used in the first place, which will then proceed to delay the attacker's application by 200-300 milliseconds. Furthermore, another consideration would be including an on-system session and a non-persistent certificate that exists on a disconnected sub-system on the processor. The Corstone-101 comes with various scripts and IP descriptions, including an

out-of-box execution test bench, verification components, IP-XACT descriptions, configuration scripts, and implementation scripts.

The system incorporates a structured approach to data handling, utilizing a predefined "Subscriber/User" data structure and a "Decoder" class for processing serialized data. A key design consideration is selecting an appropriate serialization format to ensure efficient data exchange and interoperability. JSON has been identified as the preferred serialization format, offering a lightweight, human-readable, and widely supported solution for structuring and transmitting data.

Basing it off Rutgers Computer Architecture bomb lab (computer science department)

During the bomb lab, you read the assembly to figure out where a jump command is written and there is a compare statement that appears before that jump command in order to actually execute that jump.

This compare statement technically appears before the normal phase if statement.

We will want to force our “hacker” to inject the correct hidden statement to enter the next phase otherwise, the bomb lab will loop itself. We can have filler statements during the phases to help guide when to inject that line. The whole bomb lab itself can be how they determine where to inject. My overall idea is to make this just a single phase of the overall break in just to enter the next phase which can be our tree or whatever etc.